

Clinical Trial Competitive Intelligence Dashboard

Analytics Consulting Project · Client: Fortune 500 Medical Device Manufacturer

*The Problem: Reviewers had to manually screen thousands of new clinical trial filings a month to catch **competitor moves and emerging market signals**, a process too slow and inconsistent to keep up with a market that changes every week.*

What I Built

- 1

Data Pipeline

Engineered a Python ingestion pipeline pulling 20,000+ public clinical trial records from a national trial registry API on a weekly refresh, automating sponsor deduplication and change-tracking across 10+ relational tables.
- 2

Two-Part Scoring Model

Designed a rule-based relevance score from sponsor, intervention, and condition signals, paired with an LLM-based importance score assessing trial novelty and market opportunity.
- 3

Validation & Guardrails

Stress-tested model outputs with sensitivity and robustness checks, and built an LLM guardrail layer that catches and blocks unsupported or overconfident scores before they reach stakeholders.
- 4

Interactive Dashboard

Packaged results into an executive summary view, a market-trend view, and a prioritized trial tracker, so reviewers can see not just which trials matter, but why.

Impact

\$2M+ Competitive intelligence value delivered	6 Key competitors tracked continuously
20K+ Trial records processed	3+ month Of manual review replaced
30% Cut in reporting costs	0 Unsupported scores reached stakeholders

Tools & Methods: Python · SQL · LLM Prompt Engineering · Rule-Based & ML Scoring · Sensitivity/Robustness Testing · Dashboard Design